PERFORMANCE SPECIFICATIONS FOR FIRE PROTECTION SCOPE OF WORK

GENERAL
THE CONTRACTOR IS TO ADHERE TO THIS PERFORMANCE SPECIFICATION & PROVIDE A DESIGN TO MEET ALL INTENTS OF THIS SPECIFICATION. THE WORK THAT IS INDICATED IN THIS DOCUMENT IS AN OVERVIEW OF THE PROJECT, INTENDED TO ESTABLISH THE SCOPE OF THE CONTRACTOR'S RESPONSIBILITIES. THE CONTRACTOR SHALL PREPARE COMPLETE WORKING DRAWINGS FOR SPRINKLER SYSTEM BUILD-OUT, WITH THEIR ENGINEER SEAL FOR APPROVAL BY THE APPROPRIATE AUTHORITIES. THE CONTRACTOR SHALL BE THE ENGINEER OF RECORD. THE FIRE PROTECTION CONTRACTOR SHALL PROVIDE COMPLETE & FULLY OPERABLE SPRINKLER SYSTEMS.

DESIGN SPRINKLER DRAWINGS ARE GIVEN AS GUIDE ONLY, & THEREFORE DO NOT RELIEVE THE CONTRACTOR FROM PROVIDING & INSTALLING ALL EQUIPMENT NECESSARY TO COMPLETE THE INSTALLATION ACCORDING TO THE ABOVE REQUIREMENTS & GENERAL INSTALLATION RULES AS SET FORTH BY UNDERWRITER'S AGENCIES, & NUMBER & SPACING OF SPRINKLER HEADS, SPACING & SIZE OF PIPE, DETAILS, SHALL BE AS REQUIRED BY THE VARIOUS AUTHORITIES.

THE FIRE PROTECTION SYSTEM SHALL BE INSTALLED IN STRICT ACCORDANCE WITH THE LOCAL FIRE CODE (CFC) & DIRECTIVES, THE NFPA #13D. THE REQUIREMENTS OF THE OWNER'S INSURANCE UNDERWRITER & THE AUTHORITY HAVING JURISDICTION (AHJ). COMPLETE SPRINKLER PROTECTION SHALL BE PROVIDED THROUGHOUT ALL BUILDING AREAS, UNLESS SPRINKLER OMISSION IS ALLOWED BY CODE OR AMENDMENTS, OR SPECIFICALLY NOTED OTHERWISE.

ALL COMPONENTS OF THE FIRE PROTECTION SPRINKLER SYSTEM SHALL BE UL LISTED AND/OR FACTORY MUTUAL GLOBAL APPROVED, & HAVE PRIOR APPROVAL OF THE AUTHORITY HAVING JURISDICTION FOR USE IN FIRE SPRINKLER SYSTEMS.

THE SCOPE OF WORK INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

A. OBTAIN AND PAY FOR ALL PERMITS, LICENSES, APPROVALS, REVIEWS & INSPECTIONS, & CONDUCT ALL UTILITY SHUT-DOWNS, TAPPING CONNECTIONS & TESTS.

B. PROVIDE HYDRAULIC DESIGN OF FIRE PROTECTION SYSTEMS, INCLUDING COORDINATED SHOP DRAWINGS AND CALCULATIONS, SUBMITTALS FOR APPROVALS TO ARCHITECT/ENGINEER, AHJ & OWNER'S INSURANCE UNDERWRITER. CALCULATIONS SHALL BE BASED ON NFPA #13D COMPLIANT 2-HYDRANT WATER FLOW TEST, LESS THAN (1) YEAR OLD. CONDUCT SUCH A TEST IF RECORD DATA IS UNAVAILABLE OR OUT OF DATE. ALL SUBMITTALS SHALL BE PREPARED BY A LICENSED PROFESSIONAL ENGINEER, LICENSED IN THE PROJECT JURISDICTION, AS REQUIRED BY THE LOCAL AHJ. THE CONTRACTORS ENGINEER SHALL PREPARE PLANS, CUT SHEETS, CALCULATIONS, ETC. AS REQUIRED. ALL DEFERRED SUBMITTALS SHALL BE SIGNED AND STAMPED BY THE PREPARING ENGINEER. THE CONTRACTOR SHALL INCLUDE ANY FEES REQUIRED FOR SUCH SUBMISSION IN HIS BID.

C. INSTALLATION OF APPROVED SYSTEMS INCLUDING ALL NECESSARY MATERIALS, PRODUCTS, COMPONENTS, DEVICES, SPECIALTIES & EQUIPMENT, LABOR, SERVICES, LOADING, UNLOADING, HOISTING, HANDLING & STORAGE OF MATERIALS, CUTTING, CORING & PATCHING OF PENETRATIONS OF GENERAL CONSTRUCTION, MADE TO INSTALL THE SPRINKLER WORK.

D. PIPE SLEEVES SET IN COORDINATED LOCATIONS, THROUGH WALLS, FLOORS & STRUCTURAL ELEMENTS, WITH CHROME PLATED OR PRIMED FINISHED STEEL ESCUTCHEONS PLATES FOR EXPOSED TO VIEW PIPES, PASSING THROUGH FINISHED WALLS, FLOORS AND CEILINGS.

E. SMOKE & FIRE STOPPING OF ALL PENETRATIONS CREATED BY THIS CONTRACT, PROVIDE A 2-PART MINIMUM, FIRE STOPPING SYSTEM, COMPLYING WITH LOCAL FIRE DEPARTMENT APPROVED, UL LISTED DETAILS, WHICH ESTABLISH A SMOKE OR FIRE RESISTANCE RATING, EQUAL TO OR GREATER THAN THE FLOOR OR WALL PENETRATED.

F. ALL REQUIRED ACCEPTANCE TESTING & DOCUMENTATION TO THE SATISFACTION OF AHJ.

G. PROVIDE TEMPORARY FIRE PROTECTION AS REQUIRED BY THE AHJ, FOR ALL REQUIRED CONSTRUCTION OPERATIONS.

H. CLEANUP, ON A DAILY BASIS, OF ALL DEBRIS ASSOCIATED WITH THE SPRINKLER INSTALLATION CONTRACT.

I. PROVIDE EQUIPMENT MANUALS, RECORD DRAWINGS, VALVE TAG SCHEDULES, TESTING CERTIFICATES AND PERSONNEL INSTRUCTION FOR EACH SYSTEM, PRIOR TO SYSTEM TURN-OVER TO THE OWNER.

J. FIRE PROTECTION CONTRACTOR SHALL PROVIDE ACCESS PANELS NECESSITATED BY THE CONFIGURATION OF HIS WORK, TO THE GENERAL CONTRACTOR FOR INSTALLATION. REFER TO GENERAL SPECIFICATION SECTIONS FOR THE ACCESS PANEL TYPES, REQUIRED TO BE PROVIDED.

K. FIRE PROTECTION CONTRACTOR SHALL PROVIDE PERMANENT FIRE EXTINGUISHERS AND CABINETS, AS REQUIRED BY NFPA #10 & FOR LOCATIONS SEE ARCHITECTURAL DRAWINGS.

L. ALL WORK PERFORMED BY THIS CONTRACTOR SHALL BE GUARANTEED IN WRITING, FOR A PERIOD OF NOT LESS THAN ONE YEAR, FROM THE DATE OF FINAL ACCEPTANCE.

DESIGN EACH SPRINKLER SYSTEM GIVING FULL CONSIDERATION TO THE HORIZONTAL & VERTICAL OBSTRUCTIONS TO SPRINKLER SPRAY PATTERNS, PRESENTED BY ARCHITECTURAL LAYOUTS OF ROOMS, DIFFERENT ADJACENT CEILING ELEVATIONS, BUILDING CONSTRUCTION, DUCTWORK & MECHANICAL EQUIPMENT, SUSPENDED LIGHTS, ELECTRICAL EQUIPMENT & CONDUIT BANKS. COORDINATE THE LOCATION & LAYOUT OF SPRINKLERS, PIPING & EQUIPMENT WITH THE DETAILED INSTALLATION DRAWINGS OF ALL OTHER TRADES.

SPRINKLER SYSTEM SHALL BE DESIGNED IN ACCORDANCE WITH NFPA 13D. LOCATE SPRINKLERS IN ARCHITECTURAL PATTERNS, CONSISTENT WITH CEILING GRIDS, LIGHTS, AIR SUPPLY DIFFUSERS, RETURN GRILLS & OTHER CEILING ELEMENTS, WHERE CEILINGS ARE PROVIDED FROM ARCHITECTURAL REFLECTED CEILING PLANS.

ALL SPRINKLER HEADS SHALL BE QUICK RESPONSE TYPE. CONCEALED HEADS TO BE FACTORY PAINTED TO MATCH CEILING COLOR. CONTRACTOR IS RESPONSIBLE TO COORDINATE WITH ARCHITECTS & PROVIDE COLOR SAMPLE TO MANUFACTURER PRIOR TO ORDERING.

HYDRAULIC DENSITY SHALL NOT BE LESS THAN 0.10 GPM PER SQ. FT. FOR LIGHT HAZARD 0.15 FOR ORDINARY HAZARD MECHANICAL/ELECTRICAL/TELEPHONE EQUIPMENT AREAS/STORAGE/ KITCHEN AREAS. AREA OF APPLICATION SHALL NOT BE LESS THAN 1500 SQ. FT. FOR LIGHT & ORDINARY HAZARD OCCUPANCIES; 2,000 SQ. FT. FOR ORDINARY HAZARD STORAGE AREAS QUICK RESPONSE SPRINKLERS SHALL BE UTILIZED IN ALL OCCUPANCIES. TOTAL COMBINED INSIDE & OUTSIDE HOSE ALLOWANCE SHALL BE 250 GPM, ADDED TO THE CALCULATIONS AT THE POINT OF CONNECTION TO THE CITY WATER SUPPLY. PROVIDE MINIMUM HYDRAULIC SAFETY CUSHION OF 5 PSI OR 10% OF STATIC WATER MAIN PRESSURE, WHICHEVER IS GREATER, TO HYDRAULIC CALCULATIONS.

THE ARRANGEMENT, POSITIONS AND CONNECTIONS OF PIPES, DRAINS, VALVES, SPRINKLERS, ETC. SHALL BE ESTABLISHED BY THE FIRE PROTECTION CONTRACTOR'S WORKING PLANS, WITH GUIDANCE FROM THE DESIGN DRAWINGS, AND SHALL BE CONFIGURED TO DRAIN FULLY, AVOIDING EXCESSIVE AUXILIARY DRAINS. SPRINKLER PIPING SHALL BE CONCEALED ABOVE ARCHITECTURAL CEILINGS WHERE CEILINGS ARE PROVIDED, UNLESS INDICATED OTHERWISE. ARRANGEMENT OF WET SPRINKLER PIPING MAY BE TREE, LOOP OR GRID SYSTEMS AS MAY BE HYDRAULICALLY ADVANTAGEOUS. BRANCH LINES SHALL BE ABOVE THE TOP OF LIGHTING FIXTURES WITHIN THE CEILING SPACE.

FURNISH AND INSTALL ALL SPRINKLER HEADS AS ARE REQUIRED BY CODE. SPRINKLER HEADS SHALL BE QUICK RESPONSE TYPE OF THE AUTOMATIC NORMAL TEMPERATURE UNLESS OTHERWISE SPECIFIED, SOLDER TYPE IN ACCORDANCE WITH NFPA BULLETIN NO 13. SPRINKLER HEADS SHOWN ON THIS PLAN ARE FOR INFORMATION. THE CONTRACTOR IS RESPONSIBLE TO PROVIDE ADDITIONAL SPRINKLER HEADS TO COMPLY WITH ALL APPLICABLE CODES.

COORDINATE SPRINKLER HEADS LOCATIONS WITH THE ARCHITECT DRAWINGS AND FIELD CONDITIONS. SPRINKLER HEADS IN FINISHED AREA SHALL BE CHROME PLATED, CONCEAL TYPE IN CEILING. SPRINKLER HEAD IN KITCHEN AREA & SERVICE AREA SHALL SEMI RECESSED TYPE. IN UNFINISHED AREA SHALL BE BRASS UPRIGHT & SIDE WALL SPRINKLER HEADS SHALL BE CHROME PLATED ESCUTCHEON PLATE.

SPRINKLER PIPING HANGERS SHALL BE EQUAL AND CONFIRM TO THE NFPA PAMPHLET NO. 13 REQUIREMENTS.

DRAWINGS ARE DIAGRAMMATIC AND INTENDED TO SHOW APPROXIMATE LOCATIONS UNLESS SPECIFICALLY DIMENSIONED. COORDINATE THE WORK WITH ALL TRADES TO AVOID INTERFERENCE. SPRINKLER WORK THAT IS INSTALLED WHICH INTERFERES WITH THE WORK OF OTHER TRADES DUE TO THE LACK OF COORDINATION SHALL BE CHANGED AT NO ADDITIONAL COST TO THE OWNER.

PROVIDE TAMPER SWITCH FOR ALL OS&Y VALVES AND CONNECT TO THE FIRE ALARM SYSTEM.

PROVIDE FLOW SWITCHES AND CONNECT TO THE FIRE ALARM SYSTEM.

PROVIDE FLUSH TYPE ALARM BELL STRONG AND WIRING.

PROVIDE FIRE DEPARTMENT SIAMESE CONNECTION IN CONFORMATION TO LOCAL FIRE DEPARTMENT.

PROVIDE BACK FLOW PREVENTER IN ACCORDANCE WITH LOCAL CODE REQUIREMENT.

CONTRACTOR SHALL PROVIDE MAIN/AUXILIARY DRAIN AND INSPECTOR'S TEST CONNECTIONS, CONSISTING OF REQUIRED SIZED PIPING AND GLOBE VALVE, AND SPECIAL DISCHARGE NOZZLE.

CONTRACTOR SHALL PROVIDE ALL REQUIRED AND APPROVED ENAMEL ON METAL IDENTIFICATION SIGNS AT CONTROL, DRAIN, TEST AND ALARM VALVES, AS REQUIRED BY NFPA PAMPHLET NO. 13, LATEST ISSUE. THE LOCAL FIRE CODE OFFICIAL SHALL BE NOTIFIED AT LEAST FORTY-EIGHT (48) HOURS PRIOR TO COVERING THE UNDERGROUND FIRE MAIN SO THAT AN INSPECTION OF THE INSTALLATION; TO INCLUDE JOINTS AND THRUST BLOCKS; MAY BE COMPLETED AT THE FIRE CODE OFFICIALS DISCRETION. SECTION: NFPA 24-10.10.1.10.10.2

FLUSHING OF THE WATER SUPPLY MAIN, TESTING, ADJUSTING AND RETESTING OF SPRINKLER SYSTEMS ACCORDING TO NFPA #13, UNTIL ACCEPTED BY THE FIRE DEPARTMENT & ARCHITECT. THE FIRE PROTECTION CONTRACTOR SHALL REMAIN RESPONSIBLE FOR THE SPRINKLER SYSTEMS UNTIL APPROVAL IS OBTAINED.

SPRINKLER LOCATIONS SHALL BE COORDINATED WITH ARCHITECTURAL REFLECTED CEILING PLANS OR ROOM PARTITION LAYOUTS, OR PROVIDED TO MEET NFPA #13 REQUIREMENTS IN UNFINISHED AREAS & EQUIPMENT ROOMS. SPRINKLERS INSTALLED IN AREAS OF ARCHITECTURAL REFLECTED CEILING PLANS SHALL BE QUICK RESPONSE, ORDINARY TEMPERATURE RATED, STANDARD ORIFICE, RECESSED & CONCEALED TYPES, CENTERED IN THE SHORT DIMENSION OF CEILING TILES, & AT QUARTER POINTS OF THE LONG DIMENSION, +/-1", TO PRESENT A BALANCED, SYMMETRICAL LAYOUT WITH ALL OTHER CEILING ELEMENTS. SPRINKLERS IN RESIDENTIAL AREA & CORRIDORS SHALL BE BRASS PENDENT, QUICK RESPONSE RESIDENTIAL TYPE WITH STANDARD ORIFICES, ORDINARY TEMPERATURE RATED, INSTALLED ON BRANCH LINES LOCATED TIGHT TO UNDERSIDE OF EXPOSED HEAVY TIMBER CONSTRUCTION.

SPRINKLER PIPE AND FITTINGS SHALL BE PROVIDED AS FOLLOWS:

A. BURIED AND ABOVE GROUND SPRINKLER PIPING AND FITTINGS SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA # 13. NO PIPE WITH A CORROSION RESISTANCE RATIO (CRR) LESS THAN (1), SHALL BE THREADED. GALVANIZED STEEL PIPE SHALL BE USED FOR DRY SYSTEMS, MAIN DRAINS, FDC PIPING AND WHERE PIPING IS EXPOSED TO EXTERIOR ATMOSPHERE CONDITIONS. SCHEDULE 40 PIPE SHALL BE USED FOR THREADED JOINTS 2" AND SMALLER, AND CUT-GROOVED ONLY FOR ALL GALVANIZED STEEL PIPE, 2-1/2" AND LARGER. SCHEDULE 10 LIGHT WALL BLACK PIPE, 2-1/2" AND LARGER, SHALL BE ROLL-GROOVED ONLY. APPROVED MECHANICAL GROOVED JOINTS, COUPLINGS, FITTINGS AND GROOVING EQUIPMENT SHALL BE OF THE SAME MANUFACTURER ONLY.

B. FLANGES AND FITTINGS SHALL BE PRESSURE RATED FOR THE MAXIMUM SYSTEM PRESSURE. FITTINGS SHALL BE UL OR FM, FOR USE IN SPRINKLER SYSTEMS. GASKETS FOR COUPLINGS USED IN DRY SYSTEMS SHALL BE LISTED FOR DRY SYSTEM USE. PROVIDE CONVENTIONAL HANGERS OR SEISMIC HANGERS AND SEISMIC SEPARATION ASSEMBLIES PER NFPA #13 REQUIREMENTS OF LOCAL AREA, AND/OR RESTRAINED HANGERS AT ARM-OVERS AND END OF MAINS, WHERE SYSTEM DISCHARGE PRESSURES EXCEED 100 PSI AT SPRINKLER ORIFICES.

C. PROVIDE NFPA # 24 COMPLYING WATER MAIN FLUSHING OF INCOMING FIRE SERVICE ENTRANCE.

PROVIDE A CERTIFICATE TO LOCAL FIRE DEPARTMENT STATING THAT ALL UNDERGROUND MAINS HAVE BEEN FLUSHED, TESTED AND CHLORINATED BEFORE CONNECTION TO THE SPRINKLER SYSTEM.

EXECUTION:
SPRINKLER SYSTEMS SHALL COMPLY WITH ALL REQUIRED AND ADVISORY PROVISIONS OF NFPA 13D THE LOCAL FIRE DEPARTMENT(CFC), OWNER'S INSURANCE UNDERWRITER, AND THE ARCHITECT. FLUSH TEST AND CHLORINATE UNDERGROUND IN COMING MAIN IN ACCORDANCE WITH NPPA AND ALL APPLICABLE CODE PRIOR TO CONNECTIONS TO THE SPRINKLER SYSTEM AND PROVIDE CERTIFICATION TO LOCAL FIRE DEPARTMENT.

ALL PIPE HANGERS, RODS AND ATTACHMENTS SHALL BE UL LISTED FOR FIRE PROTECTION SYSTEMS. ALL THREAD ROD SHALL NOT BE BENT. ALL PIPE HANGERS AND SUPPORTS SHALL BE INSTALLED FROM GENERAL BUILDING CONSTRUCTION ONLY, AND SYSTEM PIPING SHALL NOT BE HUNG FROM DUCTWORK, EQUIPMENT, OTHER PIPING OR PIPING SUPPORTS. ALL HANGERS AND LOCATION SHALL BE IN ACCORDANCE WITH NPPA 13.

ALL UNDERGROUND PIPING SHALL BE TESTED AT 200 PSI FLOW REQUIRED OF 1560 GPM WITH VELOCITY OF 10 FT/SEC.

PROVIDE TWO (2) HOUR HYDROSTATIC TEST (SECTION: NFPA 24-10.1.2.2) ADDITIONALLY THE FIRE CODE OFFICIAL SHALL BE NOTIFIED AT LEAST FORTY EIGHT (48) PRIOR TO WITNESS THE HYDROSTATIC TEST.

USE APPROVED FITTINGS TO MAKE CHANGES IN DIRECTION, BRANCH TAKE-OFFS FROM MAINS AND REDUCTIONS IN PIPE SIZES. BUSHINGS SHALL NOT BE USED.

INSTALL UNIONS ADJACENT TO VALVES IN PIPING 2" AND SMALLER. USE FLANGES OR FLANGE ADAPTERS IN PIPING AT VALVES OR DEVICES, 2-1/2" AND LARGER.

INSTALL SPRINKLER SYSTEM PIPING STRAIGHT AND TRUE TO BEAR EVENLY ON ALL PIPE HANGERS.

PREPARE AND INSTALL PIPING JOINTS, COUPLINGS, FITTINGS, SPECIALTY PRODUCTS AND DEVICES, ACCORDING TO THE MANUFACTURER'S WRITTEN INSTRUCTIONS.

INSTALL SPRINKLERS CENTERED IN SQUARE CEILING TILES, AND CENTERED IN THE NARROW DIMENSION, AT ANY QUARTER POINT OF RECTANGULAR CEILING TILES.

SPRINKLER DEFLECTOR POSITIONS SHALL BE PARALLEL CEILING OR TO THE SLOPE OF ROOFS OR VAULTED CEILINGS.

THE FIRE PROTECTION PIPING SHALL BE TESTED UNDER A HYDROSTATIC PRESSURE OF NOT LESS THAT 200 POUNDS, PSIG, FOR A DURATION OF NOT LESS THAN TWO HOURS AND IN ACCORDANCE WITH ALL CODE REQUIREMENTS. TO BE WITNESSES BY THE FIRE DEPARTMENT.

CONTRACTOR SHALL PREPARE SHOP DRAWINGS AND SUBMIT TO THE LOCAL FIRE PREVENTION BUREAU, INSURANCE UNDERWRITERS AND THE ARCHITECT/ENGINEER FOR REVIEW PRIOR TO INSTALLATION OF SPRINKLER SYSTEM. CONTRACTOR IS RESPONSIBLE TO PROVIDE AS-BUILT DRAWINGS.

ACCEPTANCE TESTS

THE CONTRACTOR SHALL PERFORM ALL REQUIRED ACCEPTANCE TESTING PER THE PROVISIONS OF THE APPLICABLE SECTIONS OF NFPA 13, INCLUDING COMPLETION OF ALL MATERIAL AND TEST CERTIFICATES, VERIFICATION OF THE SUCCESSFUL OPERATION OF ALL ALARM DEVICES IN THE PRESENCE OF THE AUTHORITY HAVING JURISDICTION, HYDROSTATIC PRESSURE TESTS AND FLUSHING OF THE INCOMING WATER SUPPLY MAIN. SYSTEMS & PORTIONS OF SYSTEMS, WHICH FAIL HYDROSTATIC PRESSURE TESTS, SHALL BE REPAIRED & RE-TESTED. ADDITIVES SUCH AS SODIUM SILICATE OR OTHER CHEMICALS SHALL NOT BE USED TO STOP LEAKS.

COMMISSIONING AND DEMONSTRATION:

THE CONTRACTOR SHALL PUT THE SPRINKLER SYSTEMS IN AN APPROVED OPERATIONAL CONDITION AND DEMONSTRATE TO THE OWNER'S MAINTENANCE STAFF THE NORMAL STATUS AND FAULT STATUS OF ALL SYSTEM COMPONENTS, FAMILIARIZE THEM WITH THE OPERATION AND MAINTENANCE MANUAL AND THE REQUIREMENTS OF NFPA 25.

WARRANTEE AND EMERGENCY SERVICE:

THE CONTRACTOR SHALL GUARANTEE IN WRITING FOR A PERIOD OF ONE YEAR, AFTER THE DATE OF FINAL ACCEPTANCE, ALL SYSTEMS INSTALLED UNDER THIS CONTRACT. THE CONTRACTOR SHALL BE RESPONSIBLE DURING THE INSTALLATION, TESTING AND WARRANTEE PERIODS, FOR ANY DAMAGE CAUSED BY HIM OR HIS SUBCONTRACTORS, CAUSED BY DEFECTS IN MATERIALS OR WORKMANSHIP. THE CONTRACTOR SHALL PROVIDE DURING THE WARRANTEE PERIOD, EMERGENCY REPAIR SERVICE FOR THE SPRINKLER SYSTEMS, WITHIN (4) FOUR HOURS OF A REQUEST BY THE OWNER FOR SUCH SERVICE. THIS SERVICE SHALL BE PROVIDED ON A 24-HOUR, 7-DAY A WEEK BASIS UNTIL THE EXPIRATION OF THE WARRANTY PERIOD.

CONTRACTOR TO FURNISH & INSTALL FIRE EXTINGUISHER IN ACCORDANCE W/NFPA STANDARD NO. 10.

FE-1 FIRE EXTINGUISHERS
10 LB ABC MULTIPURPOSE EXTINGUISHERS WITH WALL BRACKET.

FE-2 FIRE EXTINGUISHERS
9 LB. CLASS K, 60B.C CARTRIDGE OPERATED DRY-CHEMICAL WITH WALL BRACKET

FEC-1 FIRE EXTINGUISHERS
10 LB. 2BC MULTIPURPOSE EXTINGUISHERS IN RECESSED CABINET.

NOTE:

OWNER TO REVIEW AND VERIFY WHETHER THEY NEED THESE NOTES

FIRE PROTECTION LEGEND

	CHECK VALVE		CAP	AFF	ABOVE FINISHED FLOOR
	GATE VALVE ON RISE		DROP DOWN	A/S	AUTOMATIC SPRINKLER
	RISE UP		FIRE DEPARTMENT CONNECTION	FL	FIRE LINE
	SPRINKLER - CONCEALED TYPE		MAIN SHUT OFF VALVE	M	METER
FS	FLOW SWITCH		INSPECTORS TEST AND DRAIN VALVE	PG	PRESSURE GAUGE
	RELIEF VALVE		OS&Y GATE VALVE	TYP	TYPICAL
	SIGHT GLASS		FIRE LINE	VIF	VERIFY IN FIELD
	AUTOMATIC AIR VENT WITH GLOBE VALVE AT ALL HIGH POINTS	P	PRESSURE GUAGE	FS	FLOW SWITCH
M	WATER METER		ANCHOR	FDC	FIRE DEPARTMENT CONNECTION
XX	NOZZLE TAG NUMBER	XX	NODE NUMBER		WALL HORNS AND STROBES

PROJECT CODE COMPLIANCE
1. FIRE PROTECTION SYSTEM SHALL BE INSTALLED WITH STRICT ACCORDANCE WITH NFPA 13D

SHEET INDEX

Sr No.	SHEET NO	DESCRIPTION	SCALE
1	FP1.0	FIRE PROTECTION SPECIFICATION	N.T.S.
2	FP2.0	FIRE SPRINKLER PLAN	1/4" = 1'-0"
3	FP3.0	FIRE SPRINKLER RISER DIAGRAM AND DETAILS	N.T.S.

Project
RESIDENCE OF
JOE HUBBARD
3495
COUNTY ROAD P
GLENN COUNTY,CA

Project No.:
DATE:
DRAWN BY:
DESIGNED BY:

Sheet Title
FIRE PROTECTION SPECIFICATION AND SYMBOL

☐ Preliminary Dwg.
☐ Bidding Doc.
☐ Construction Doc.

Sheet:
FP1.0

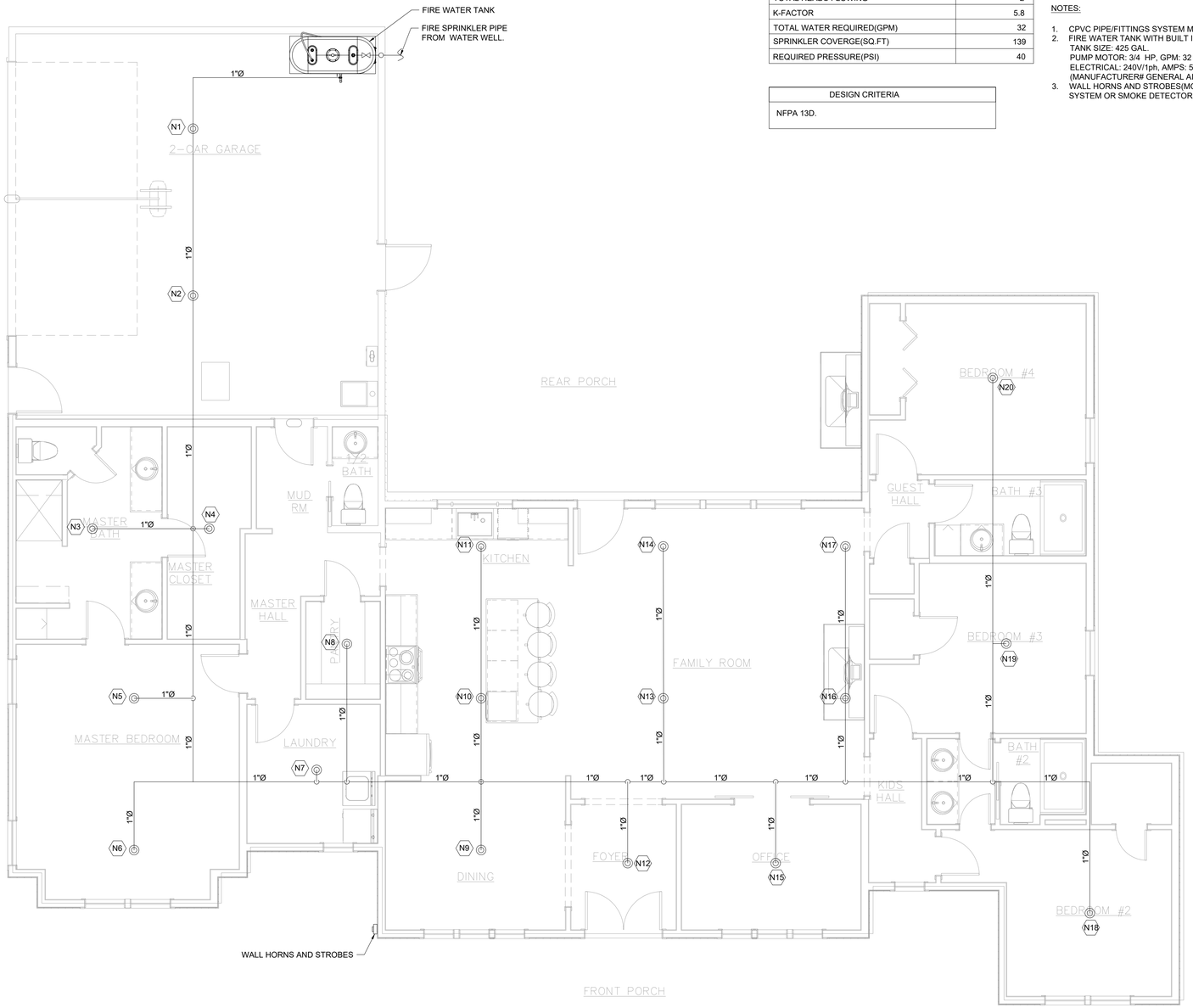
TABLE-1:BASIS OF DESIGN FOR DWELLING UNIT	
REMOTE AREA #1(2 HEAD CALC)	
OCCUPANCY CLASS	LIGHT HAZARD
TYPE OF SPRINKLER	RESIDENTIAL
TOTAL HEADS FLOWING	2
K-FACTOR	5.8
TOTAL WATER REQUIRED(GPM)	32
SPRINKLER COVERGE(SQ.FT)	139
REQUIRED PRESSURE(PSI)	40

DESIGN CRITERIA
NFPA 13D.

MIN WATER / PUMP REQUIREMENTS
A MINIMUM OF 40 PSI AT 32 GPM AT THE BOTTOM OF THE FIRE SPRINKLER RISER IS REQUIRED. IF ANY LESS WATER IS PROVIDED,IT MAY RENDER THE SYSTEMS UNUSABLE.

NOTES:

1. CPVC PIPE/FITTINGS SYSTEM MUST BE LISTED BY A THIRD PARTY TO THE UL 1821 STANDARD.
2. FIRE WATER TANK WITH BUILT IN PUMP.
   TANK SIZE: 425 GAL.
   PUMP MOTOR: 3/4 HP, GPM: 32 AND HEAD: 15FT.
   ELECTRICAL: 240V/1ph, AMPS: 5.2.
   (MANUFACTURER# GENERAL AIR PRODUCT, MODEL# H2H425-E01).
3. WALL HORNS AND STROBES(MODEL NO# EG1F-HDV1575) TO BE CONNECTED TO SECURITY SYSTEM OR SMOKE DETECTOR CIRCUIT TO ARCHIVE AUDIBILITY IN SLEEPING AREA.



1 FP2.0 FIRE SPRINKLER FIRST FLOOR PLAN

SCALE: 1/4" = 1'-0"

NORTH

Project

RESIDENCE OF

JOE HUBBARD

3495
COUNTY ROAD P
GLENN COUNTY,CA

Project No.:

DATE:

DRAWN BY:

DESIGNED BY:

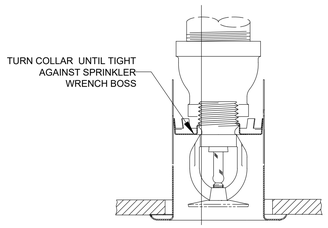
Sheet Title

FIRE SPRINKLER ZONING PLAN

☐ Preliminary Dwg.
☐ Bidding Doc.
☐ Construction Doc.

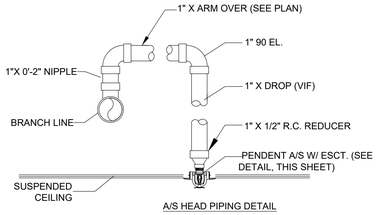
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FP2.0



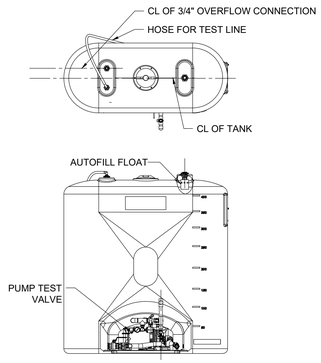
A/S HEAD MOUNTING DETAIL

1 PENDENT TYPE FIRE SPRINKLER DETAIL
SCALE:



A/S HEAD PIPING DETAIL

2 FIRE SPRINKLER DETAIL
SCALE:



3 FIRE TANK BUIT IN PUMP DETAIL
SCALE:

SPRINKLER SYSTEM DESIGN CRITERIA								
AREA SERVED	MFR. & MODEL NO	HAZARD CLASS	SPRINKLER DENSITY (GPM)	AREA OF APPLICATION (SQ.FT)	ORIFICE SIZE & K FACTOR	CONFIGURATION	TEMPERATURE RATING	REMARKS
ALL AREA	VIKING-QUICK RESPONSE-VK498	LIGHT HAZARD	0.1	3000	1/2" K=5.8	PENDENT,CONCEALED TYPE W/ COVER PLATE (SPACE PERMITTING)	155° F	WHITE FINISH COVER

NOTES:
1. SMALL THERMOSENSITIVE, GLASS BULB RESIDENTIAL SPRINKLER.
2. MAXIMUM WORKING PRESSURE IS 175 PSI (12 BAR).

Project
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3495
COUNTY ROAD P
GLENN COUNTY,CA

Project No.:
DATE:
DRAWN BY:
DESIGNED BY:

Sheet Title
FIRE SPRINKLER DETAILS AND SCHEDULE

☐ Preliminary Dwg.
☐ Bidding Doc.
☐ Construction Doc.

Sheet:
FP3.0